PLSC 508: Homework 1

Visualization, Centrality, Transitivity/Reciprocity, & Community Detection

Jody Park

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Instruction

Find a network dataset that has been used in a previously published paper, but is not an example dataset from a software package, and has not been used in an in-class tutorial (other than your own, if you have given yours). Apply the following methods, present the results in the form of a figure or table, and interpret what you find.

1. Visualize the network.
2. Calculate and analyze at least one form of centrality.
3. Calculate and analyze transitivity and/or reciprocity on the network.
4. Perform community detection.

Submit a PDF write-up, as well as the data and code required to reproduce your findings.

GitHub Repository

Code and data for reproducing this file are available in my GitHub repository. The repository README explains how to download the large data files from Harvard Dataverse.

Data and network construction

```
# read the original bureaucratic witness dyads
hw_data <- haven::read_dta("bureaucrats_jop/dyad_b.dta")
stopifnot(!anyDuplicated(hw_data$dyad_id),
          !anyNA(hw_data[c("asker", "agencycode", "congress")]),
          all(nzchar(hw_data$agencycode)))
hw_congress <- 113

# reproduce Table 2 appointment and presidential alignment coding
hw_data <- hw_data %>%
  mutate(pas = as.numeric(polappt_type == "PAS"),
         pas = ifelse(agencycode == "DD", NA_real_, pas),
         pas = ifelse(!is.na(dod_pol), dod_pol, pas),
         outparty = 1 - pres_party,
         appointment = case_when(
           pas == 1 ~ "PAS / Defense political",
```

```
          polappt == 0 | (agencycode == "DD" & dod_pol == 0) ~ "Career / Defense
↪  nonpolitical",
          TRUE ~ "Other / unknown"),
        party = case_when(party_code == 100 ~ "Democrat",
                          party_code == 200 ~ "Republican",
                          TRUE ~ "Other"))
hw_sample <- hw_data %>% filter(congress == hw_congress)

# a tie records participation; weight counts witness-member-hearing rows
hw_edges <- hw_sample %>%
  group_by(asker, agencycode) %>%
  summarise(weight = n(), .groups = "drop")
hw_members <- hw_sample %>% distinct(asker, party)
stopifnot(!anyDuplicated(hw_members$asker))
hw_members <- hw_members %>% arrange(asker)
hw_agencies <- sort(unique(hw_sample$agencycode))
hw_B <- matrix(0, nrow(hw_members), length(hw_agencies),
               dimnames = list(paste0("M", hw_members$asker), hw_agencies))
hw_B[cbind(match(hw_edges$asker, hw_members$asker),
           match(hw_edges$agencycode, hw_agencies))] <- hw_edges$weight
hw_g <- graph_from_biadjacency_matrix(hw_B, weighted = TRUE)
hw_binary <- 1L * (hw_B > 0)

# the member projection counts shared agencies, not direct member exchanges
hw_shared <- tcrossprod(hw_binary)
diag(hw_shared) <- 0
hw_project <- graph_from_adjacency_matrix(hw_shared, mode = "undirected",
                                          weighted = TRUE, diag = FALSE)
stopifnot(is_bipartite(hw_g), !is_directed(hw_g),
          ecount(hw_g) == nrow(hw_edges))

# white tables follow the existing draft style
hw_table <- function(x, caption, digits = 3) {
  knitr::kable(x, format = "latex", booktabs = TRUE, longtable = TRUE,
               digits = digits, caption = caption, row.names = FALSE) %>%
    kableExtra::kable_styling(full_width = FALSE, position = "center",
                              font_size = 9, latex_options = "repeat_header",
                              repeat_header_method = "replace") %>%
    kableExtra::row_spec(0:nrow(x), background = "white", color = "black")
```

```
}
hw_pal <- c(Democrat = "#0072B2", Republican = "#D55E00",
            Other = "#999999", Agency = "#009E73")
```

```
# describe the observed network, not an eligible committee roster
hw_description <- data.frame(
  Congress = hw_congress, Members = nrow(hw_B), Agencies = ncol(hw_B),
  Ties = sum(hw_binary), Rows = nrow(hw_sample),
  Hearings = n_distinct(hw_sample$hearingid),
  Density = mean(hw_binary))
hw_table(hw_description, "Observed participation in the 113th Congress")
```

Table 1: Observed participation in the 113th Congress

Congress	Members	Agencies	Ties	Rows	Hearings	Density
113	419	51	2854	10998	831	0.134

Table 1 describes the network I analyze. The data come from Ban et al. (2026). Each row in `dyad_b.dta` records a bureaucratic witness and a member who interacted in a House committee hearing. The file contains 61,802 rows from 6,686 hearings in the 105th through 113th Congresses (1997–2014).

I analyze the 113th Congress (2013–2014), the last Congress in the archive. Membership changes from one Congress to the next. Pooling all years would join members who never served together, so I keep a single Congress as the main network.

The **network** has two modes. Members form one mode and agency codes form the other. A **tie** joins member m and agency a when m questioned a witness from a at least once. The 113th Congress network contains 419 members and 51 agency codes with 2,854 ties. The tie weight counts witness–member–hearing rows. It does not count individual questions or the amount of analytical information. I keep the analytical score as a separate variable because contact frequency and content measure different things (Ban et al., 2026; Liu et al., 2025). Legislative and oversight hearings both enter the network. Section 2 compares their content.

Density equals 0.134. It is the share of member–agency pairs with a tie among members and agencies that appear in the data. A zero means that the data record no exchange. It does not show that the member had a chance to question that agency. Committee rosters would help define this opportunity set. Without them, I cannot separate an absent tie from an absent opportunity. All results below therefore describe observed participation rather than access. I also retain the source `agencycode`. The `DD` code pools Defense headquarters with the military departments. I assign no agency ideology scores in this assignment.

1. Network Visualization

Bipartite structure and shared agency contacts

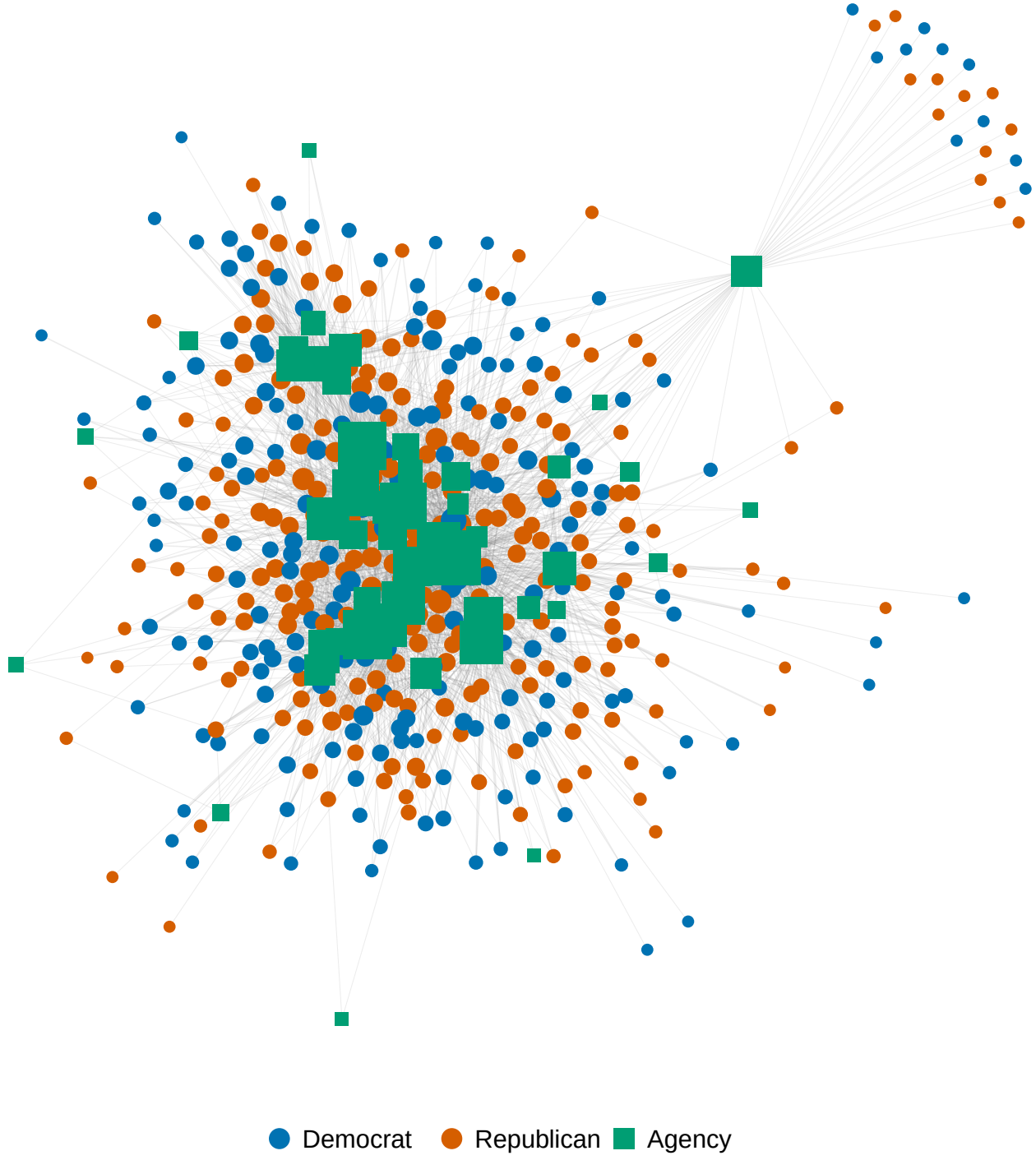
I visualize the 113th Congress network in three ways. Figures 1 and 2 draw the same graph with two layouts. The force-directed layout pulls together nodes with shared contacts. The two-column layout separates members from agencies and orders members by party and then degree, and agencies by degree. Both figures use the same encodings. Circles are members colored by party, and squares are agencies. Node size increases with degree. In the two-column layout, members share one small size and agency size is capped. Labeled sidebars identify the two party blocks in the dense member column. Edge width increases with the number of witness–member–hearing rows. Figure 3 removes the edges and plots the incidence matrix directly. Brighter cells mark more frequent contact on a log scale, and dark cells mark no recorded exchange. Layout distances are not ideological distances.

```
# shared encodings for both layouts: party color, mode shape, degree size
set.seed(508)
hw_xy <- layout_with_fr(hw_g, weights = NA)
hw_type <- V(hw_g)$type
hw_colors <- c(unname(hw_pal[hw_members$party]),
               rep(hw_pal["Agency"], ncol(hw_B)))
hw_size <- 1.8 + 0.6 * sqrt(degree(hw_g))
hw_shape <- ifelse(hw_type, "square", "circle")
hw_edge_width <- 0.2 + log1p(E(hw_g)$weight) / 3

# one-line legend centered below the network
hw_legend <- function() {
  legend("bottom", c("Democrat", "Republican", "Agency"),
         col = hw_pal[c("Democrat", "Republican", "Agency")],
         pch = c(16, 16, 15), pt.cex = 2, cex = 1.1, horiz = TRUE,
         bty = "n", inset = c(0, -0.07), xpd = NA)
}

op <- par(mar = c(3, 1, 1, 1))
plot(hw_g, layout = hw_xy, vertex.label = NA, vertex.size = hw_size,
     vertex.color = hw_colors, vertex.frame.color = NA, vertex.shape = hw_shape,
     edge.color = adjustcolor("gray50", alpha.f = 0.15),
     edge.width = hw_edge_width)
hw_legend()
```

Figure 1: Observed legislator–agency exchanges in the 113th Congress: force-directed layout




```
par(op)
```

Figure 1 shows a dense core of large departments. Members of both parties surround this core without a visible partisan split. Contact breadth is uneven. The median member questioned witnesses from 6 agencies, and the most active member questioned 30. Agencies vary more. The median agency faced 40 members. The Department of Defense (DD) faced 226, or 54% of observed members.

Two kinds of nodes sit outside the core. One agency sits apart with a fan of members attached only to it. This agency is the Office of the U.S. Trade Representative (TN). Section 2 traces this fan to a single committee. Small commissions and boards sit at the edge with few contacts. 6 agency codes faced four or fewer members. These peripheral agencies reach a narrow audience. These degrees alone do not show whether committee jurisdiction accounts for their narrow audiences.

```
# members: Democrats above Republicans, then by degree; agencies by degree
hw_order_m <- order(hw_members$party, -rowSums(hw_binary))
hw_order_a <- order(-colSums(hw_binary))
hw_y_m <- hw_y_a <- NULL
hw_y_m[hw_order_m] <- seq(1, -1, length.out = nrow(hw_B))
hw_y_a[hw_order_a] <- seq(1, -1, length.out = ncol(hw_B))
hw_two <- cbind(as.numeric(hw_type), c(hw_y_m, hw_y_a))

# party sidebars make the dense member column easy to distinguish
op <- par(mar = c(3, 1, 2, 1))
plot(hw_g, layout = hw_two, vertex.label = NA,
     xlim = c(-1.7, 1.7), ylim = c(-1.05, 1.1), asp = 0,
     vertex.size = ifelse(hw_type,
                          pmin(0.5 + 0.1 * sqrt(degree(hw_g)), 2.3), 1.0),
     vertex.color = hw_colors, vertex.frame.color = NA, vertex.shape = hw_shape,
     edge.color = adjustcolor("gray50", alpha.f = 0.07),
     edge.width = hw_edge_width)
text(c(-1, 1), 1.1, c("Members", "Agencies"), font = 2, cex = 1.2, xpd = NA)
# label each party block independently of tiny individual nodes
for (party_name in c("Democrat", "Republican")) {
  party_y <- hw_y_m[hw_members$party == party_name]
  segments(-1.10, min(party_y), -1.10, max(party_y),
           col = hw_pal[party_name], lwd = 6, lend = "butt")
  text(-1.40, mean(range(party_y)),
       paste0(party_name, "s\n(n = ", length(party_y), ")"),
       col = hw_pal[party_name], font = 2, cex = 0.95)
}
# match the party sidebars with the observed agency count
segments(1.10, min(hw_y_a), 1.10, max(hw_y_a),
```

```
         col = hw_pal["Agency"], lwd = 6, lend = "butt")
text(1.40, mean(range(hw_y_a)),
     paste0("Agencies\n(n = ", ncol(hw_B), ")"),
     col = hw_pal["Agency"], font = 2, cex = 0.95)
hw_legend()
par(op)
```

Figure 2: Observed legislator–agency exchanges in the 113th Congress: two-column layout ordered by party and degree

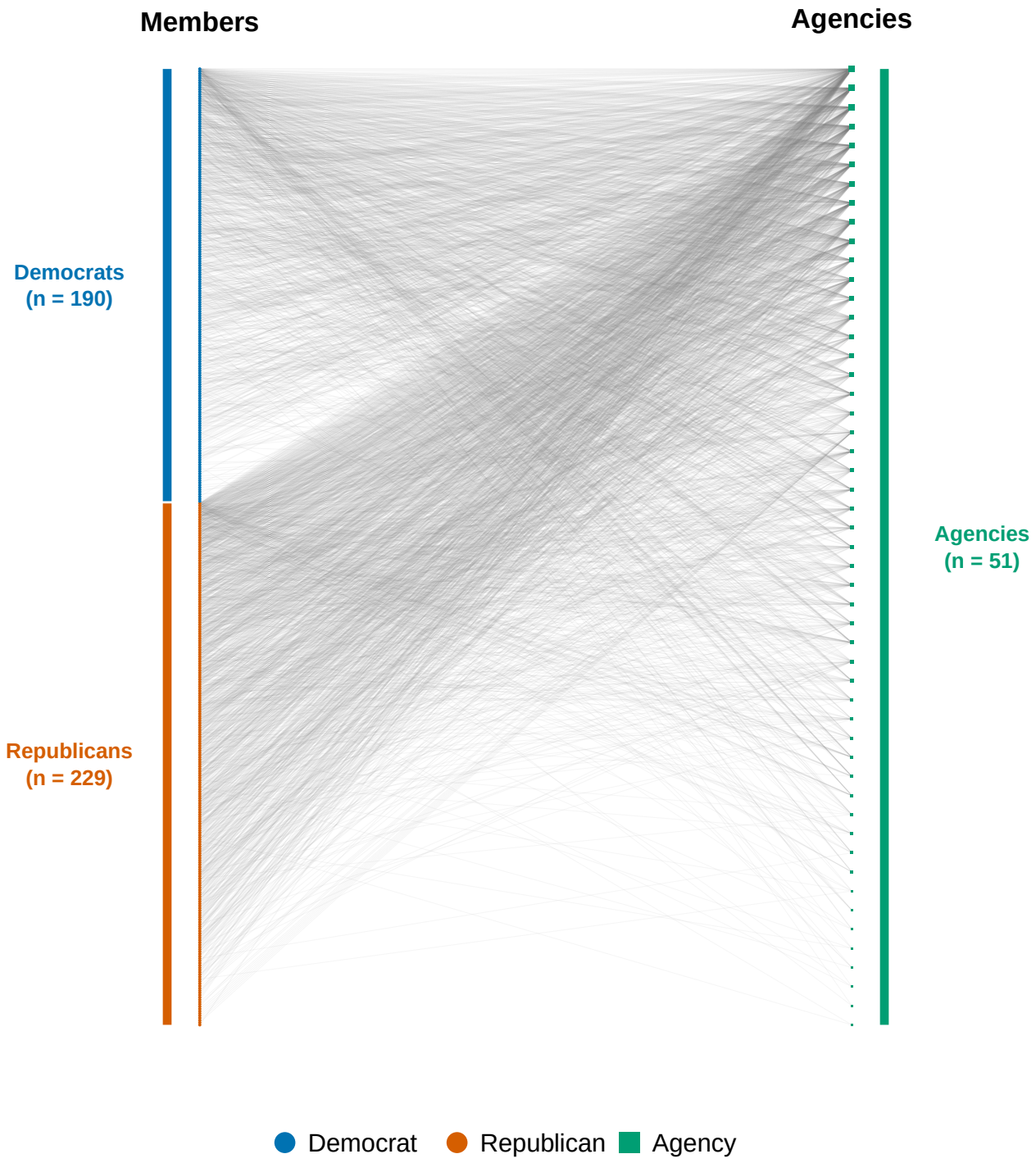


Figure 2 preserves the bipartite structure. The blue sidebar identifies Democrats, the orange sidebar identifies Republicans, and the green sidebar identifies agencies. Each sidebar reports the number of observed nodes in its group. The agency column shows the distribution of contacts across these party blocks. The agencies with the highest degree sit at the top, and their edges fan out to both the Democratic block and the Republican block. Edge density thins quickly further down the column. Most agencies draw a few members from a single region of the member column. Individual contacts are hard to trace because edges cross often. The figure shows how unequal agency degree is rather than who questioned whom.

```
# show contact counts with a labeled log-scale color key
hw_heat <- expand.grid(Member = seq_len(nrow(hw_B)),
                       Agency = seq_len(ncol(hw_B)))
hw_heat$Rows <- as.vector(hw_B[hw_order_m, hw_order_a])
ggplot(hw_heat, aes(Member, Agency, fill = Rows)) +
  geom_raster() +
  geom_vline(xintercept = sum(hw_members$party == "Democrat") + 0.5,
             color = "white", linetype = "dashed", linewidth = 0.3) +
  scale_fill_viridis_c(trans = "log1p", breaks = c(0, 1, 5, 20, 100),
                       name = "Dyad rows (log scale)") +
  scale_x_continuous(expand = c(0, 0)) +
  scale_y_continuous(expand = c(0, 0)) +
  labs(x = "Members (Democrats, then Republicans)",
       y = "Agencies (decreasing degree)") +
  theme_minimal(base_size = 12) +
  theme(legend.position = "bottom", panel.grid = element_blank()) +
  guides(fill = guide_colorbar(title.position = "top", title.hjust = 0.5,
                               barwidth = grid::unit(9, "cm")))
```

Figure 3: Member-by-agency incidence: members ordered by party and degree; agencies ordered by degree

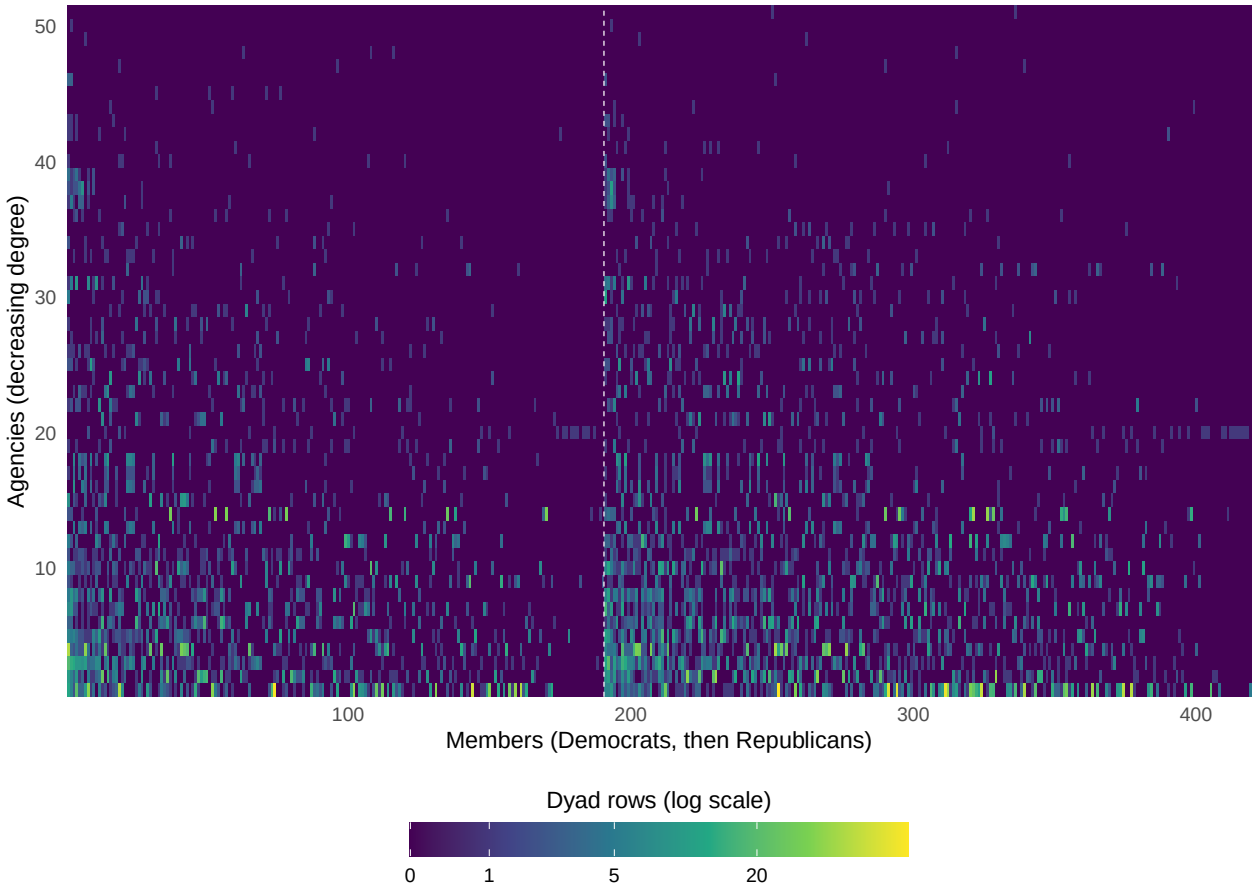


Figure 3 shows every member–agency pair without edge crossings. Bright cells cluster in the bottom rows, which hold the agencies with the highest degree. These agencies draw members on both sides of the dashed party line. Rows above them are sparse. Most agencies reach a small set of members. Within each party, members on the left of the block contact many agencies and members on the right contact few. The breadth of contact varies more within parties than between them.

These patterns describe observed participation in a shared venue. In Ban et al. (2026), hearings are the setting in which bureaucrats share information with members. The figures show which agencies reach many members in that setting. They do not show that members communicate with each other. Two members tied to the same agency belong to its observed audience. They need not have attended the same hearing or questioned the same witness.

2. Centrality

Contact breadth, repeated participation, and network position

I calculate four measures on the bipartite graph. Degree k_i counts the distinct partners of node i. Degree share divides degree by the size of the opposite mode:

$$\text{Degree share}_i = \frac{k_i}{n_{\text{opposite}}}$$

The denominator counts observed nodes only. It is not a share of eligible partners. Strength sums the tie weights of node i, so it counts repeated witness–member–hearing rows. Eigenvector centrality gives a node a high score when its partners also have high scores. Betweenness measures how often node i lies on shortest paths between other nodes:

$$B_i = \frac{2}{(n-1)(n-2)} \sum_{s<t,\ s,t\neq i} \frac{\sigma_{st}(i)}{\sigma_{st}}$$

Here σ_{st} is the number of shortest paths between nodes s and t. The term $\sigma_{st}(i)$ counts those that pass through i. The first fraction rescales the sum by the number of node pairs. I compute eigenvector centrality and betweenness on the binary graph. Degree and eigenvector centrality are radial measures because they count walks that start or end at a node. Betweenness is medial because it counts paths that pass through a node (Borgatti & Everett, 2006). None of these measures shows that information actually traveled along a path. I then compare members by party and turn from contact to the content of each exchange.

```
# binary paths describe contact structure; frequency is a separate strength
hw_central <- data.frame(
  Node = V(hw_g)$name,
  Mode = ifelse(hw_type, "Agency", "Member"),
  Degree = degree(hw_g),
  DegreeShare = degree(hw_g) / ifelse(hw_type, nrow(hw_B), ncol(hw_B)),
  Strength = strength(hw_g),
  Eigenvector = eigen_centrality(hw_g, weights = NA)$vector,
  Betweenness = betweenness(hw_g, weights = NA, normalized = TRUE))
hw_central$Party <- c(hw_members$party, rep("Agency", ncol(hw_B)))

# top nodes within each mode; do not rank the two modes together
hw_top <- hw_central %>% group_by(Mode) %>%
  arrange(desc(Degree), Node, .by_group = TRUE) %>% slice_head(n = 5) %>%
  ungroup() %>% select(-Party)
hw_rank_cor <- cor(hw_central$Degree[hw_type],
                   hw_central$Betweenness[hw_type], method = "spearman")
```

```r
# members whose only observed agency contact is the trade representative
hw_tn_members <- hw_members$asker[hw_binary[, "TN"] == 1 & rowSums(hw_binary) ==
 ↪  1]
hw_tn_only <- length(hw_tn_members)
hw_tn_wm <- n_distinct(hw_sample$asker[hw_sample$asker %in% hw_tn_members &
                         hw_sample$committee_name == "Committee on Ways and
 ↪  Means"])
hw_tn_dem <- sum(hw_members$party[hw_members$asker %in% hw_tn_members] ==
 ↪  "Democrat")

# committee service of the three members with the broadest contact
hw_broad <- hw_members$asker[order(-rowSums(hw_binary))][1:3]
hw_broad_ogr <- n_distinct(hw_sample$asker[hw_sample$asker %in% hw_broad &
  hw_sample$committee_name == "Committee on Oversight and Government Reform"])
```

```r
# report four measures, with the opposite-mode degree denominator
hw_table(hw_top, "Top five nodes per mode by degree; binary path measures")
```

Table 2: Top five nodes per mode by degree; binary path measures

Node	Mode	Degree	DegreeShare	Strength	Eigenvector	Betweenness
DD	Agency	226	0.539	2151	1.000	0.213
HE	Agency	173	0.413	952	0.850	0.114
TR	Agency	165	0.394	600	0.795	0.111
ST	Agency	155	0.370	1046	0.752	0.092
HS	Agency	151	0.360	520	0.772	0.083
M412272	Member	30	0.588	159	0.366	0.012
M400273	Member	26	0.510	110	0.337	0.008
M400090	Member	24	0.471	87	0.314	0.010
M400196	Member	23	0.451	112	0.316	0.007
M412482	Member	23	0.451	77	0.326	0.007

Table 2 lists the five nodes with the highest degree in each mode. I rank members and agencies separately because their denominators differ. `DD` leads on every agency measure. It reached 54% of observed members and accounts for 2,151 rows. The next four agencies are also cabinet departments. Strength reorders them. The State Department (`ST`) ranks fourth in degree but second in strength. Its strength records more witness–member–hearing rows despite its lower degree.

The member rows show a different kind of outlier. The most active member reached 59% of agency codes. All 3 of the three broadest members questioned witnesses on the Committee on Oversight

and Government Reform. That committee can investigate any agency. Its members therefore meet witnesses from across the executive branch. This pattern is consistent with broad committee jurisdiction; the measures do not separate jurisdiction from individual effort. Member betweenness stays small. Every path between members passes through at least one agency.

```
# label the five agencies with the most member contacts and the trade outlier
hw_agency_central <- hw_central %>% filter(Mode == "Agency")
ggplot(hw_agency_central, aes(DegreeShare, Betweenness)) +
  geom_point(aes(size = Strength), color = hw_pal["Agency"], alpha = 0.7) +
  geom_text(data = hw_agency_central %>% filter(rank(-Degree) <= 5 | Node ==
   ↪  "TN"),
            aes(label = Node), nudge_y = 0.002, size = 4) +
  scale_size_continuous(range = c(2, 9)) +
  labs(x = "Share of observed members contacted", y = "Normalized betweenness",
       size = "Dyad rows") +
  theme_minimal(base_size = 12) +
  theme(legend.position = "bottom", panel.grid.minor = element_blank()) +
  guides(size = guide_legend(nrow = 1))
```

Figure 4: Agency contact breadth and binary betweenness in the bipartite graph

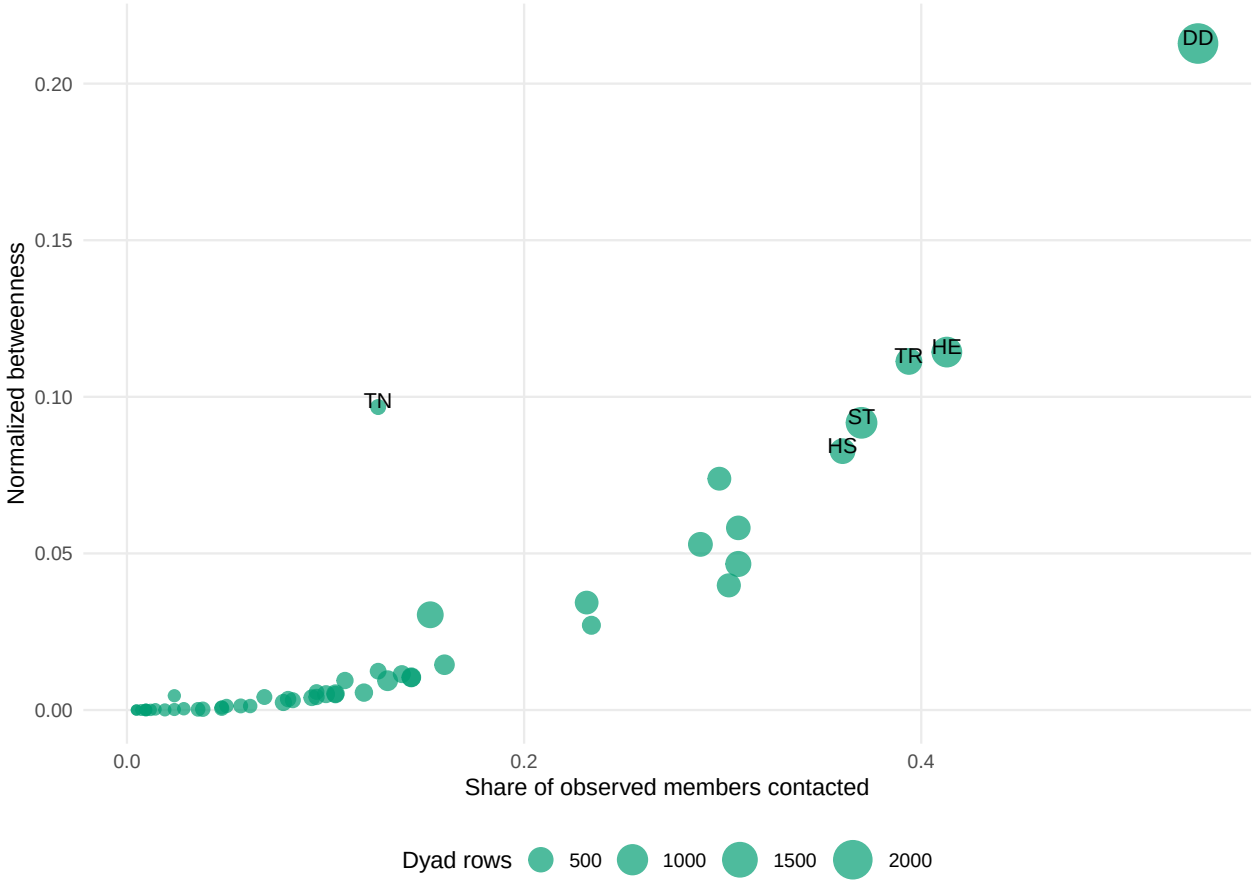


Figure 4 compares agency degree share with betweenness.[1] The two measures move together. Their Spearman rank correlation is 0.971. Agencies that reach many members also connect many otherwise distant members. `TN` is the clear exception. It ranks 4th in betweenness with only 53 member contacts. Of these members, 22 questioned no other agency. All 22 of them questioned witnesses on the Committee on Ways and Means, which holds jurisdiction over trade. The group is bipartisan, with 10 Democrats and 12 Republicans. Every shortest path from them to the rest of the graph passes through `TN`. Its medial position reflects a separate committee audience rather than broad contact.

```
# compare members only, avoiding different denominators across modes
hw_party_central <- hw_central %>% filter(Mode == "Member") %>%
  group_by(Party) %>% summarise(N = n(), Degree = mean(Degree),
    Strength = mean(Strength), Eigenvector = mean(Eigenvector),
    Betweenness = mean(Betweenness), .groups = "drop")
```

[1]TN: Office of the U.S. Trade Representative; HS: Department of Homeland Security; ST: Department of State; TR: Department of the Treasury; HE: Department of Health and Human Services; DD: Department of Defense, including the military departments.

```
hw_table(hw_party_central, "Mean member centrality by party")
```

Table 3: Mean member centrality by party

Party	N	Degree	Strength	Eigenvector	Betweenness
Democrat	190	6.405	23.905	0.104	0.001
Republican	229	7.148	28.192	0.118	0.001

Table 3 compares members by party. Republicans have a higher mean degree than Democrats (7.15 versus 6.41). Their mean strength is also higher. The differences are small. They describe observed participation rather than a party effect. Majority status and committee jurisdiction could produce the same gap. Republicans held the House majority in the 113th Congress.[2]

```
# appointment is an exchange attribute, not a fixed agency attribute
hw_content <- hw_sample %>%
  mutate(Hearing = ifelse(oversight == 1, "Oversight", "Legislative"),
         Alignment = ifelse(outparty == 1, "Out-partisan", "Co-partisan")) %>%
  group_by(Hearing, appointment, Alignment) %>%
  summarise(Rows = n(), Members = n_distinct(asker),
            Agencies = n_distinct(agencycode),
            Score = mean(ascore, na.rm = TRUE), .groups = "drop")
hw_table(hw_content, "Analytical content by hearing, appointment, and
 ↪  presidential alignment",
         digits = 2)
```

Table 4: Analytical content by hearing, appointment, and presidential alignment

Hearing	appointment	Alignment	Rows	Members	Agencies	Score
Legislative	Career / Defense nonpolitical	Co-partisan	1120	153	38	33.12
Legislative	Career / Defense nonpolitical	Out-partisan	1568	196	39	30.91
Legislative	Other / unknown	Co-partisan	114	63	14	34.29
Legislative	Other / unknown	Out-partisan	167	75	14	28.33
Legislative	PAS / Defense political	Co-partisan	421	128	27	33.92
Legislative	PAS / Defense political	Out-partisan	580	164	27	30.49
Oversight	Career / Defense nonpolitical	Co-partisan	1734	169	41	33.76
Oversight	Career / Defense nonpolitical	Out-partisan	2556	208	40	31.36
Oversight	Other / unknown	Co-partisan	125	63	16	36.51
Oversight	Other / unknown	Out-partisan	223	93	17	32.40
Oversight	PAS / Defense political	Co-partisan	1028	165	33	34.76
Oversight	PAS / Defense political	Out-partisan	1362	203	34	31.26

[2]U.S. House of Representatives, 113th Congress profile.

```r
# co-partisan minus out-partisan mean within each hearing and appointment group
hw_gap <- hw_content %>% group_by(Hearing, appointment) %>%
  summarise(Gap = Score[Alignment == "Co-partisan"] -
              Score[Alignment == "Out-partisan"], .groups = "drop")
```

Table 4 turns from contact to content. `Score` averages the exchange-level analytical scores from Ban et al. (2026) on their 0–100 scale. Each exchange-level score summarizes the analytical content of a witness's testimony to a member in a hearing. `Alignment` records whether the member shares the President's party. Appointment status describes the witness in each row. It is not a fixed agency attribute, so one agency can appear in several groups. I follow the PAS coding and Defense adjustment in Ban et al. (2026). Unlike their binary regression indicator, this descriptive table separates other or unknown appointment types from identified career officials. PAS denotes presidential appointees confirmed by the Senate. The table covers 10,998 rows. Group means range from 28.33 to 36.51.

Out-partisans of the President receive lower mean scores in every hearing-by-appointment group. The gap ranges from 2.21 to 5.95 points. It is larger for PAS appointees than for career officials in both hearing types. In oversight hearings the PAS gap is 3.50 points and the career gap is 2.40. These raw means agree in direction with Ban et al. (2026). They do not reproduce the published estimates, which include fixed effects and controls.

Centrality and content answer different questions. Centrality describes who reaches whom. It does not describe what passes through a tie (Liu et al., 2025). `DD` has the broadest audience in the network. Its breadth says nothing about how much analysis its witnesses provide.

3. Transitivity & Reciprocity

Projection closure and bipartite four-cycles

Reciprocity is not defined for this network. Reciprocity asks whether a tie from i to j is matched by a tie from j to i. That question requires directed ties. Each tie here records a joint exchange in a hearing. I could direct each tie from the questioning member to the agency. Every reverse tie would then be zero by construction. I could instead add each witness answer as a reverse tie. Every tie would then be reciprocated by construction. Neither choice tests reciprocity, so I analyze transitivity.

Ordinary triangle transitivity is uninformative in the bipartite graph. Members connect only to agencies. A triangle would need a tie between two members or between two agencies. The bipartite graph therefore contains no triangles. This zero follows from the structure of the graph; it is not evidence of weak cohesion (Cranmer et al., 2021; Lo et al., 2026). Therefore, I use two alternatives. The first projects the network onto members, so two members share a tie when they questioned a common agency. The second counts four-cycles in the bipartite graph itself.

Global transitivity in the member projection is the share of connected triples that close into triangles:

$$C = \frac{3 \times \text{number of triangles}}{\text{number of connected triples}}$$

Mean local clustering averages closure across members:

$$\bar{C} = \frac{1}{n} \sum_{i=1}^{n} \frac{t_i}{k_i(k_i - 1)/2}$$

Here t_i counts the triangles that include member i and k_i is the projected degree. Members with $k_i < 2$ receive zero. The global measure weights members by their number of triples. The local mean weights every member equally (Tam Cho & Fowler, 2010). A four-cycle joins two members through two shared agencies. Let s_{ij} be the number of agencies that members i and j both questioned. The count is

$$F = \sum_{i<j} \binom{s_{ij}}{2}$$

It is a count rather than a proportion. I compare all three statistics with a random reference and then repeat them for each Congress.

```r
# calculate three nontrivial closure summaries from the incidence matrix
hw_closure <- function(B) {
  shared <- tcrossprod(B)
  common <- shared[upper.tri(shared)]
  diag(shared) <- 0
  g <- graph_from_adjacency_matrix(1L * (shared > 0),
                                   mode = "undirected", diag = FALSE)
  c(Global = transitivity(g, type = "global", isolates = "zero"),
    MeanLocal = transitivity(g, type = "average", isolates = "zero"),
    FourCycles = sum(common * (common - 1) / 2))
}
hw_observed <- hw_closure(hw_binary)
hw_bipartite_triangles <- sum(count_triangles(hw_g)) / 3
stopifnot(hw_bipartite_triangles == 0)

# randomize agency choices, retaining every member's observed degree
# this reference does not preserve agency degree or committee opportunities
set.seed(508)
hw_draws <- 99
hw_null <- replicate(hw_draws, {
  B <- matrix(0L, nrow(hw_binary), ncol(hw_binary))
  for (i in seq_len(nrow(B))) {
    B[i, sample.int(ncol(B), rowSums(hw_binary)[i])] <- 1L
  }
  hw_closure(B)
})
hw_closure_table <- data.frame(
  Measure = names(hw_observed), Observed = unname(hw_observed),
  NullMean = rowMeans(hw_null),
  NullLow = apply(hw_null, 1, quantile, probs = 0.025),
  NullHigh = apply(hw_null, 1, quantile, probs = 0.975))
```

```r
# compare all three statistics against the same bipartite reference
# four-cycles are counts, so they print as integers
hw_table(hw_closure_table %>%
           mutate(across(-Measure, ~ ifelse(Measure == "FourCycles",
                                            format(round(.x), big.mark = ","),
                                            sprintf("%.3f", .x)))),
         "Closure versus 99 row-degree-preserving random incidence matrices")
```

Table 5: Closure versus 99 row-degree-preserving random incidence matrices

Measure	Observed	NullMean	NullLow	NullHigh
Global	0.845	0.680	0.677	0.683
MeanLocal	0.870	0.737	0.734	0.739
FourCycles	209,952	60,252	59,497	61,357

Table 5 reports the observed values next to the reference. `Observed` is the value in the 113th Congress network. `NullMean` is the mean across random draws, and `NullLow` and `NullHigh` bound the middle 95% of draws. Global transitivity is 0.845 and mean local clustering is 0.870. Both values are high. Much of this closure follows from the projection. An agency questioned by three members creates a triangle among them. `DD` alone joins 226 members into one clique (Lo et al., 2026). The projection has a density of 0.703. The bipartite graph contains 209,952 four-cycles.

Each of the 99 random draws keeps every member's observed degree and assigns agencies at random. The reference keeps both mode sizes. It does not keep agency degree or committee opportunities, and some agencies can receive no members in a draw. The reference means are 0.680 for global transitivity and 0.737 for mean local clustering. The mean four-cycle count is 60,252. All three observed values exceed the 97.5th percentile of their reference distributions. Members share agencies more often than random choice predicts. The four-cycle gap is the most informative of the three. Projection inflates both triangle measures in the observed and the random networks alike. Four-cycles instead count repeated overlap, where two members share more than one agency.

The comparison does not explain why. Agency popularity is one candidate. A few departments draw many members, and shared audiences follow. Committee jurisdiction is another, since members of one committee question the same agencies. The Ways and Means members who questioned only `TN` illustrate this mechanism. The reference removes both features. It is an illustrative benchmark rather than a test conditioned on committee rosters. It also says nothing about whether ideology produces closure.

```
# changing coverage means this is a descriptive comparison across Congresses
hw_by_congress <- bind_rows(lapply(sort(unique(hw_data$congress)), function(t) {
  d <- hw_data %>% filter(congress == t) %>% distinct(asker, agencycode)
  B <- 1L * (table(d$asker, d$agencycode) > 0)
  z <- hw_closure(B)
  data.frame(Congress = t, Members = nrow(B), Agencies = ncol(B),
             Global = z[1], MeanLocal = z[2], FourCycles = z[3])
}))
hw_table(hw_by_congress, "Closure in observed networks across Congresses")
```

Table 6: Closure in observed networks across Congresses

Congress	Members	Agencies	Global	MeanLocal	FourCycles
105	134	32	0.709	0.811	2684
106	260	45	0.765	0.830	39732
107	268	43	0.746	0.810	28341
108	324	41	0.803	0.844	95439
109	349	53	0.789	0.830	73074
110	421	51	0.837	0.861	172740
111	422	57	0.856	0.874	179884
112	421	52	0.872	0.885	272397
113	419	51	0.845	0.870	209952

Table 6 repeats the observed calculation for each Congress. Global transitivity rises from 0.709 in the 105th Congress to 0.872 in the 112th. The observed network grows from 134 to 421 members over the same period. Changes in transcript coverage can alter measured closure, but larger networks do not necessarily have higher transitivity. Transcript coverage and membership change across these years. I therefore treat the trend as descriptive.

For Ban et al. (2026), hearings matter because several members hear the same bureaucrats at once. High closure means that these audiences overlap. Members who share agencies may question different witnesses in different hearings. Four-cycles establish repeated overlap in agency contacts, not repeated exposure to the same witnesses. That overlap does not show that they cooperate.

4. Community Detection

Two algorithms on the weighted member projection

I detect communities in the weighted member projection. The edge weight counts the agencies that two members share. I use two algorithms. Louvain searches for a partition with high weighted modularity. Walktrap merges members whose short random walks of four steps end in similar places (Yang et al., 2016). Modularity compares the weight inside communities with the weight expected from member strengths:

$$Q = \frac{1}{2W} \sum_{i,j} \left[w_{ij} - \frac{s_i s_j}{2W} \right] \delta(c_i, c_j)$$

Here w_{ij} is the shared-agency weight between members i and j. The strength s_i sums the weights of member i, and W is the total edge weight. The indicator $\delta(c_i, c_j)$ equals one when the two members belong to the same community. I compare the two partitions with the adjusted Rand index. The index equals one for identical partitions and zero for agreement expected by chance. I then compare the Louvain partition with party and with each member's modal committee.

```
# use shared-agency counts as similarities for both algorithms
set.seed(508)
hw_louvain <- cluster_louvain(hw_project, weights = E(hw_project)$weight)
hw_walktrap <- cluster_walktrap(hw_project, weights = E(hw_project)$weight,
                                steps = 4)
hw_communities <- list(Louvain = hw_louvain, Walktrap = hw_walktrap)
hw_modularity <- data.frame(
  Algorithm = names(hw_communities),
  Groups = sapply(hw_communities, function(x) length(unique(membership(x)))),
  Modularity = sapply(hw_communities, function(x)
    modularity(hw_project, membership(x), weights = E(hw_project)$weight)))
hw_agreement <- compare(hw_louvain, hw_walktrap, method = "adjusted.rand")

# compare the same partition with party and modal observed committee
hw_modal_committee <- hw_sample %>% count(asker, committee_name) %>%
  arrange(asker, desc(n), committee_name) %>% group_by(asker) %>%
  slice_head(n = 1) %>% ungroup()
hw_member_groups <- hw_members %>%
  mutate(Community = as.integer(membership(hw_louvain)),
         Committee = hw_modal_committee$committee_name[
           match(asker, hw_modal_committee$asker)])
hw_party_ari <- compare(hw_member_groups$Community,
                        as.integer(factor(hw_member_groups$party)),
```

```
                        method = "adjusted.rand")
hw_committee_ari <- compare(hw_member_groups$Community,
                            as.integer(factor(hw_member_groups$Committee)),
                            method = "adjusted.rand")
hw_composition <- hw_member_groups %>% group_by(Community) %>%
  summarise(Members = n(), Democrats = sum(party == "Democrat"),
            Republicans = sum(party == "Republican"),
            TopCommittee = names(which.max(table(Committee))),
            .groups = "drop") %>%
  mutate(TopCommittee = sub("^Committee on (the )?", "", TopCommittee))

# identify each community's most frequently contacted agency code
hw_agency_groups <- hw_edges %>%
  left_join(hw_member_groups %>% select(asker, Community), by = "asker") %>%
  group_by(Community, agencycode) %>% summarise(Contacts = n(), .groups = "drop")
  ↪ %>%
  arrange(Community, desc(Contacts), agencycode) %>% group_by(Community) %>%
  slice_head(n = 1) %>% ungroup() %>%
  rename(TopAgency = agencycode)
hw_composition <- left_join(hw_composition, hw_agency_groups, by = "Community")
```

```
# compare modularity on the identical weighted graph
hw_table(hw_modularity, "Community detection on shared agency contacts")
```

Table 7: Community detection on shared agency contacts

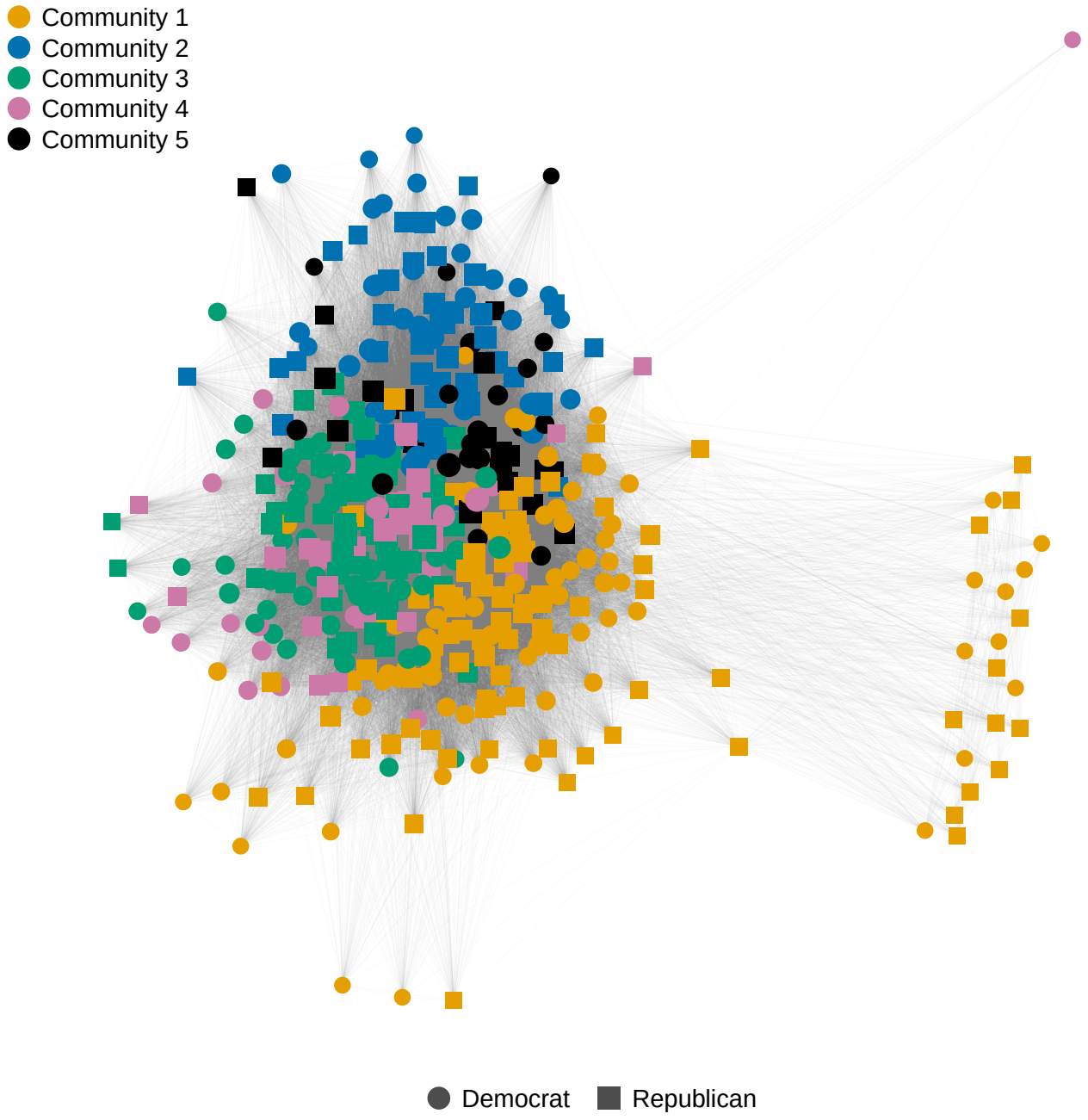
Algorithm	Groups	Modularity
Louvain	5	0.114
Walktrap	6	0.083

Table 7 reports the results. Louvain finds 5 communities with $Q = 0.114$. Walktrap finds 6 with $Q = 0.083$. Both values are modest. The projection is dense, so most members share an agency with most other members. The partitions also disagree. Their adjusted Rand index is 0.179. I use Louvain as the main partition. Its higher modularity does not make it the true partition.

```
# Okabe-Ito colors separate communities; shape carries party
set.seed(508)
hw_project_xy <- layout_with_fr(hw_project, weights = E(hw_project)$weight)
hw_group_colors <- c("#E69F00", "#0072B2", "#009E73", "#CC79A7",
                     "#000000", "#56B4E9", "#D55E00")[
```

```
                    seq_len(max(hw_member_groups$Community))]
op <- par(mar = c(3, 1, 1, 1))
plot(hw_project, layout = hw_project_xy, vertex.label = NA,
     vertex.size = 3 + sqrt(rowSums(hw_binary)) / 2,
     vertex.color = hw_group_colors[hw_member_groups$Community],
     vertex.frame.color = NA,
     vertex.shape = ifelse(hw_members$party == "Democrat", "circle", "square"),
     edge.color = adjustcolor("gray50", alpha.f = 0.04),
     edge.width = 0.15 + log1p(E(hw_project)$weight) / 5)
legend("topleft", paste("Community", seq_along(hw_group_colors)),
       col = hw_group_colors, pch = 16, pt.cex = 2.2, cex = 1.1, bty = "n")
legend("bottom", c("Democrat", "Republican"), pch = c(16, 15),
       col = "gray30", pt.cex = 2.2, cex = 1.1, horiz = TRUE,
       bty = "n", inset = c(0, -0.07), xpd = NA)
```

Figure 5: Louvain communities among members connected through shared agencies




```
par(op)
```

Figure 5 shows the 5 Louvain communities. Color marks the community and shape marks the party. Node size increases with the number of agencies a member questioned, and edge width increases with the number of shared agencies. The communities occupy different regions of the layout but overlap at the center. Circles and squares mix within every color, so party does not separate the groups visually. The `TN` audience appears as a separate cluster on the right. Louvain places it in

the community led by DD. This partition places the small, tight cluster within a larger group; it does not by itself establish why the algorithm grouped them together.

```
# counts reveal whether algorithmic groups simply reproduce party
hw_table(hw_composition %>% rename(Agency = TopAgency, Committee = TopCommittee),
         "Louvain community composition, leading agency, and modal committee", 0)
```

Table 8: Louvain community composition, leading agency, and modal committee

Community	Members	Democrats	Republicans	Committee	Agency	Contacts
1	149	64	85	Armed Services	DD	104
2	79	38	41	Financial Services	TR	72
3	99	47	52	Energy and Commerce	DN	90
4	51	22	29	Oversight and Government Reform	HE	47
5	41	19	22	Judiciary	HS	39

Table 8 describes each community. `Agency` is the agency code contacted by the largest number of distinct members in the community. `Contacts` reports that number, not the number of exchanges. `Committee` is the most common modal committee among community members. Party does not organize the partition. Every community contains Democrats and Republicans in similar proportions. The adjusted Rand index with party is −0.002. Agencies and committees track the partition more closely. Each community has a different leading agency, and all five are cabinet departments. The adjusted Rand index with each member's modal observed committee is 0.205. The modal committee is the committee in which a member appears most often in these data. I break ties alphabetically. It is not the member's full committee assignment. The agreement is higher than for party but still limited. The most common modal committee usually fits the leading agency. Armed Services leads the DD community and Financial Services leads the TR community.

These communities partition members into nonoverlapping groups whose agency audiences can overlap. They are not legislative alliances or groups of witnesses (Ban et al., 2026; Leifeld & Schneider, 2012). Both algorithms use a projection and ordinary modularity. Neither estimates communities in the bipartite graph itself. Projection can exaggerate community structure because each shared agency creates a clique (Lo et al., 2026). The low agreement between algorithms also limits claims about a unique partition.

5. Conclusion

The 113th Congress hearing network shows stronger descriptive alignment with committees than with party. The visualizations show a core of cabinet departments that members of both parties question. Centrality confirms this core. `DD` reached 54% of observed members, while `TN` gained high betweenness through a separate Ways and Means audience. Party differences in member centrality are small, but co-partisans of the President receive more analytical content in every group. Transitivity in the member projection is high and exceeds a random reference, although projection inflates it. Communities have modest modularity ($Q = 0.114$) and follow committees and leading agencies more than party.

These results help me shape the extension paper. Access and content should be modeled separately. The partisan gap in Ban et al. (2026) appears in content, but observed participation is bipartisan and communities align more closely with committees than with party. Committee rosters would define which member–agency pairs could form, so absent ties would carry meaning. Models that work on the bipartite graph directly would avoid the closure and community inflation that projection creates (Lo et al., 2026). An interaction between member party and agency ideology could then measure alignment for each member–agency pair to test whether it is associated with analytical content.

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